QUESTION 1: CALCULATED NUMERIC

10 out of 10 points

4-bit quantization is applied on the signal x(t)=16cos(200π t+π/7). If the mean-square-error can be written as MSE=a/12, what is the value of the constant a?

Given Answer: 4

Correct Answer: 4

Answer range +/- 0 (4 - 4)

QUESTION 2: MULTIPLE CHOICE

10 out of 10 points

3-bit quantization is applied on the signal $x(t) = \cos(200\pi t) + 3\cos(600\pi t)$ with the encoding rule shown below:

Interval	Codeword
[3 4]	111
[2 3]	110
[1 2]	101
[0 1]	100
[-1 0]	011
[-2 -1]	010
[-3 -2]	001
[-4 -3]	000

If x(t) is sampled at 1000 Hz, what is the binary sequence corresponding to the sample x[3]?

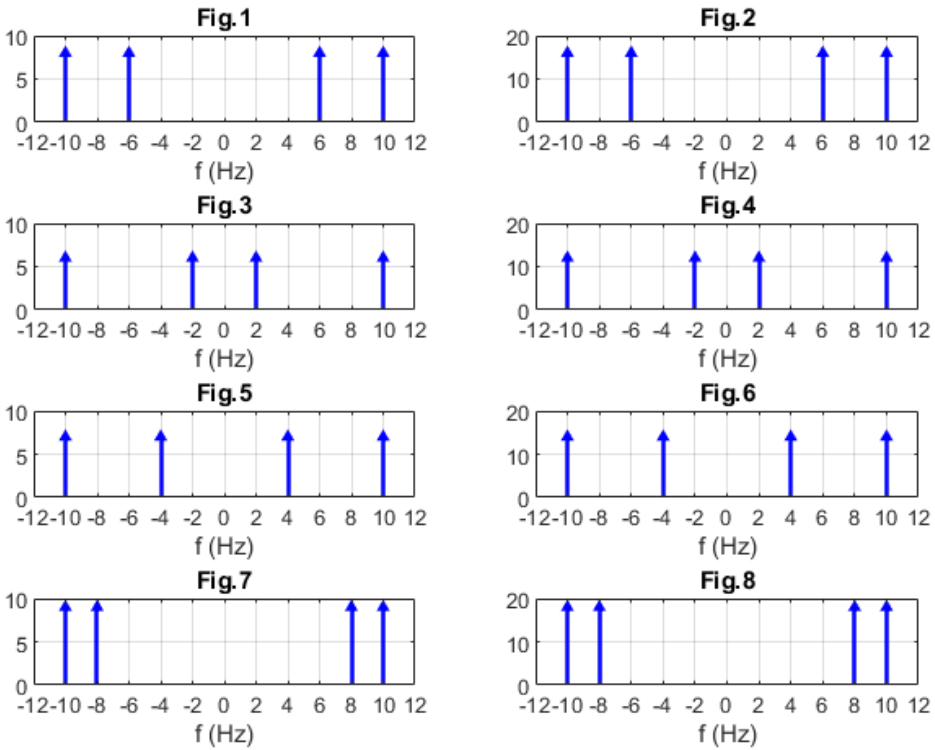
Given Answer: 110

Correct Answer: 110

QUESTION 3: MULTIPLE CHOICE

15 out of 15 points

The signal $x(t) = \cos(20\pi t)$ is sampled at 0.9 times the Nyquist rate. Which one of the below figures corresponds to the spectrum of the sampled signal (between -12 Hz and 12 Hz)?



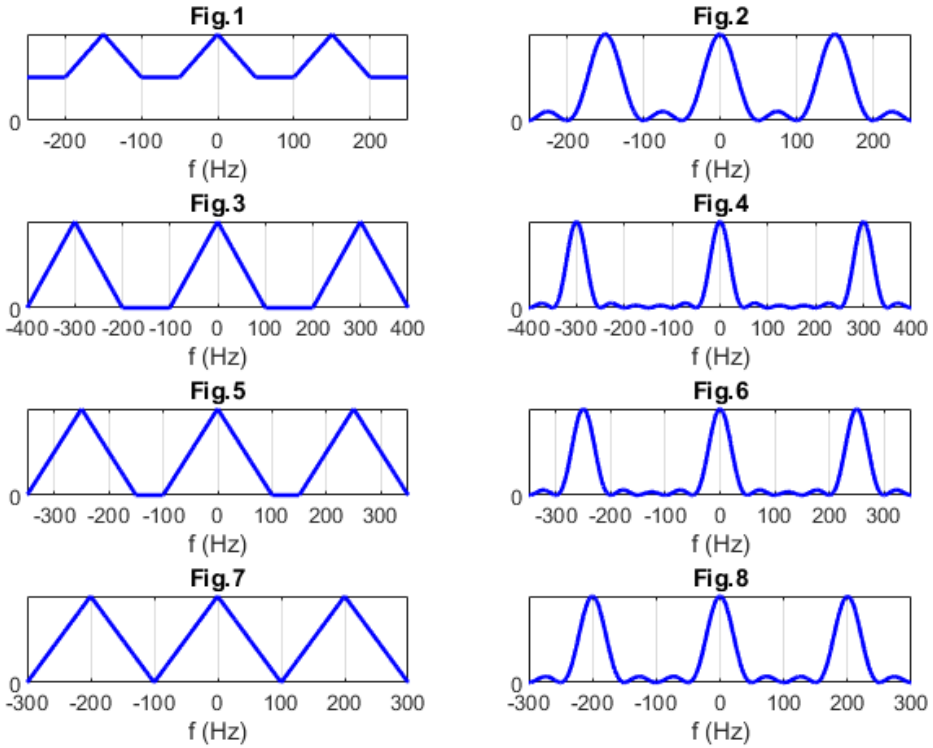
Given Answer: Fig. 7

Correct Answer: Fig. 7

QUESTION 4: MULTIPLE CHOICE

10 out of 10 points

The signal $x(t)=\text{sinc}^2(100t)$ is sampled at 1.25 times the Nyquist rate. Which one of the below figures corresponds to the spectrum of the sampled signal?



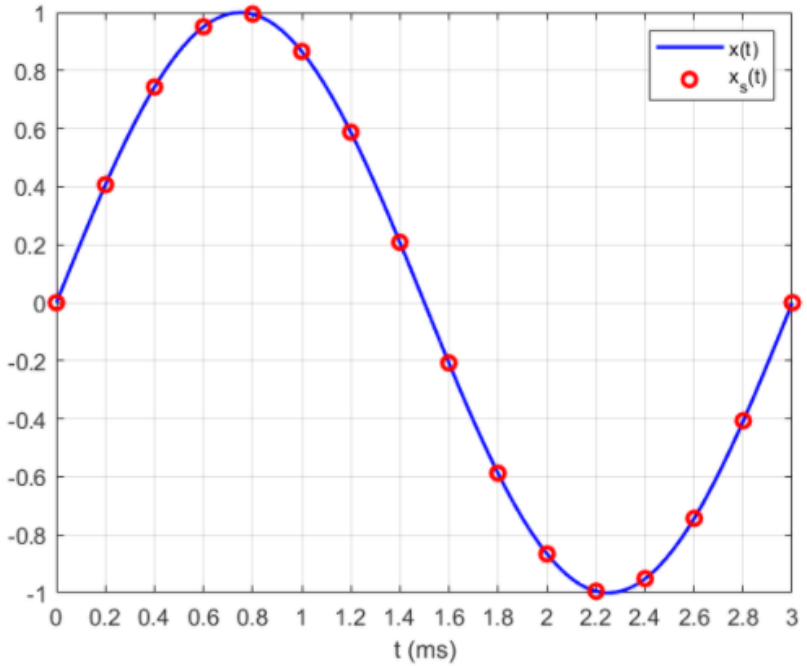
Given Answer: Fig. 5

Correct Answer: Fig. 5

QUESTION 5: CALCULATED NUMERIC

15 out of 15 points

The below figure shows one cycle of a sinusoidal signal x(t) as well as the sampled signal $x_s(t)$. If x(t) is sampled at k times the Nyquist rate, what is the value of k?



Given Answer: 7.5

Correct Answer: 7.5

Answer range +/- 0 (7.5 - 7.5)

QUESTION 6: CALCULATED NUMERIC

10 out of 10 points

An analog-to-digital converter (ADC) is used to digitize 10 seconds of a music signal where it is assumed that the sampler of the ADC operates at a fixed sampling frequency. With 32-level quantization, the digital file occupies a storage space of 100 kbits. Find the value of the storage space (in kbits) if 512-level quantization is performed.

Given Answer: 180

Correct Answer: 180

Answer range +/- 0 (180 - 180)

QUESTION 7: MULTIPLE CHOICE

10 out of 10 points

Consider the following signals. Which one of the signals needs to be sampled at the lowest rate?

$x(t) = \text{sinc}^2(200t)$

$y(t) = \text{sinc}^2(100t)$

$z(t) = \text{sinc}(200t)$

$w(t) = \text{sinc}(100t)$

Given Answer: w(t)

Correct Answer: w(t)

QUESTION 8: CALCULATED NUMERIC

10 out of 10 points

Consider a signal whose Fourier Transform is given by: X(f)=rect(f/10)+rect(f/20).
What is the value of the sampling frequency (in Hz) if the signal is sampled at 5 times the Nyquist rate?

Given Answer: 100

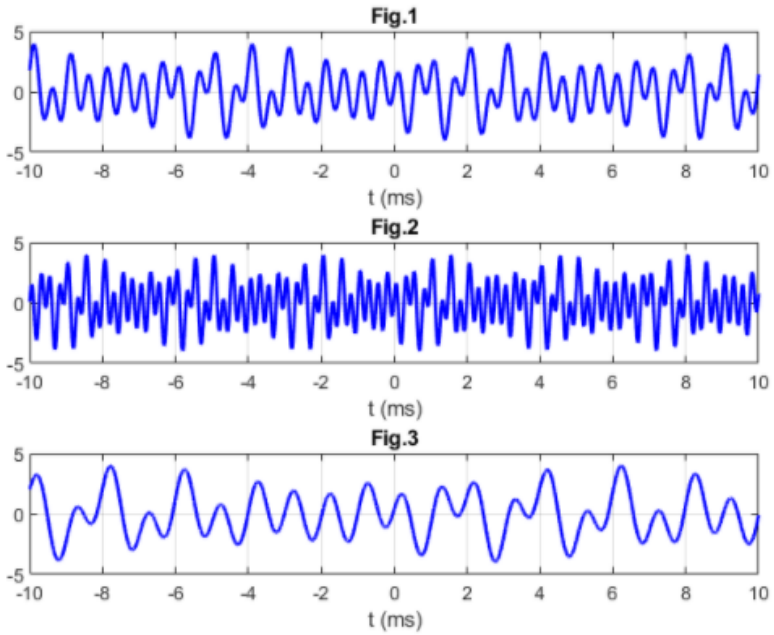
Correct Answer: 100

Answer range +/- 0 (100 - 100)

QUESTION 9: MULTIPLE CHOICE

10 out of 10 points

Consider the signals shown below. Which one of the below figures corresponds to the signal that needs to be sampled at the lowest rate?



Given Answer: Fig. 3

Correct Answer: Fig. 3